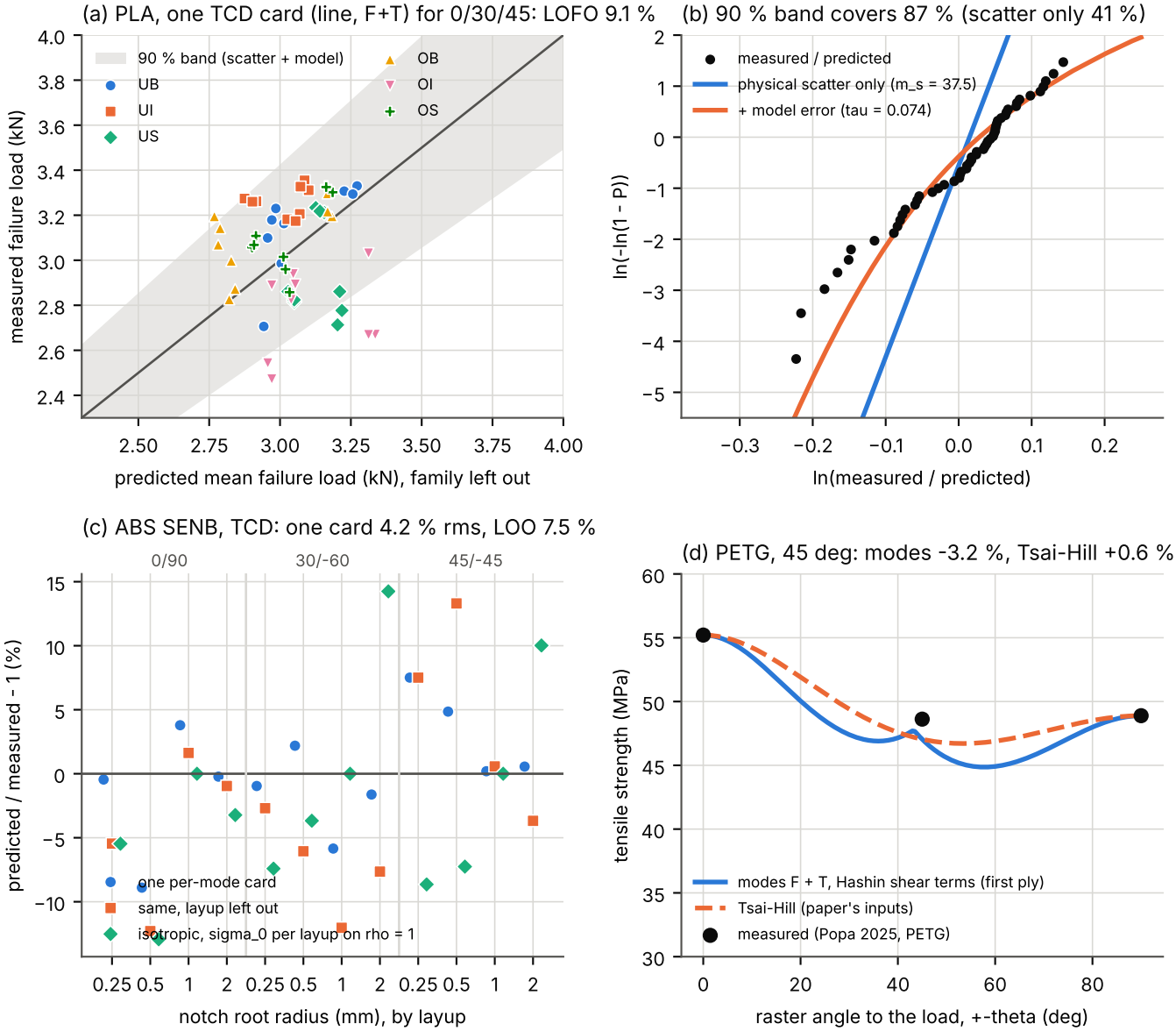
(a) PLA, one TCD card (line, F+T) for 0/30/45: LOFO 9.1 %
90 % band (scatter + model)
UB
UI
US
OB
OI
OS
measured failure load (kN)
predicted mean failure load (kN), family left out
(b) 90 % band covers 87 % (scatter only 41 %)
measured / predicted
physical scatter only (m_s = 37.5)
+ model error (tau = 0.074)
ln(-ln(1 - P))
ln(measured / predicted)
(c) ABS SENB, TCD: one card 4.2 % rms, LOO 7.5 %
0/90
30/-60
45/-45
predicted / measured - 1 (%)
one per-mode card
same, layup left out
isotropic, sigma_0 per layup on rho = 1
notch root radius (mm), by layup
(d) PETG, 45 deg: modes -3.2 %, Tsai-Hill +0.6 %
tensile strength (MPa)
modes F + T, Hashin shear terms (first ply)
Tsai-Hill (paper's inputs)
measured (Popa 2025, PETG)
raster angle to the load, +-theta (deg)